



A pre-feasibility study for the establishment of a center for the collection and marketing of sheep milk

Page 1



2017

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1. Executive Summary

The project aims to collect sheep milk in Karak governorate from the sheep breeders and in collection center, then distributing and marketing the milk to the dairy factories in the various governorates due to the increase of inability of sheep breeders to market their products of fresh milk in a timely manner and spoiled.

The problem of marketing milk in a timely manner, especially in the season of milking is one of the main problems facing sheep and goat breeders in remote and desert areas

Project Name	Collection and marketing of sheep milk Center
The Proposed Site	Al-Karak
The Products of the Project	Supply of sheep and goats milk
Total Labor	9
Total Investment (,000 JD)	387
Net Present Value of Cash Flows (,000 JD)	919
Payback Period (Years)	6
Internal Rate of Return	%25.64

The idea of the project is to collect sheep's milk from different areas in Karak by using refrigerated transport means , collection, processing and marketing to major dairy factories inside and outside the province and dairy workshops and cheese and Jameed producers.

The estimated capital cost of the project is JD 386,724, mostly for refrigerated transport vehicles and transport vehicles (121,000 JD) and working capital (108,314 JD). The working capital is required to purchase milk from breeders in cash and to sell to dairy factories by post payment. (50%) of the project will be financed through loans and the rest through self financing. Despite the high investment cost of the project, it will provide suitable employment opportunities commensurate with the size of the investment because of the the project depends on employment in the process of milk collection and marketing.

The results of the financial analysis indicate that the net t value of the cash flows of the project is positive, which is an indicator of the financial feasibility of the project and the internal rate of return is much higher than the weighted average cost of capital, which is an indicator of the feasibility of the project. Other financial indicators point that the project is profitable to the investor.

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

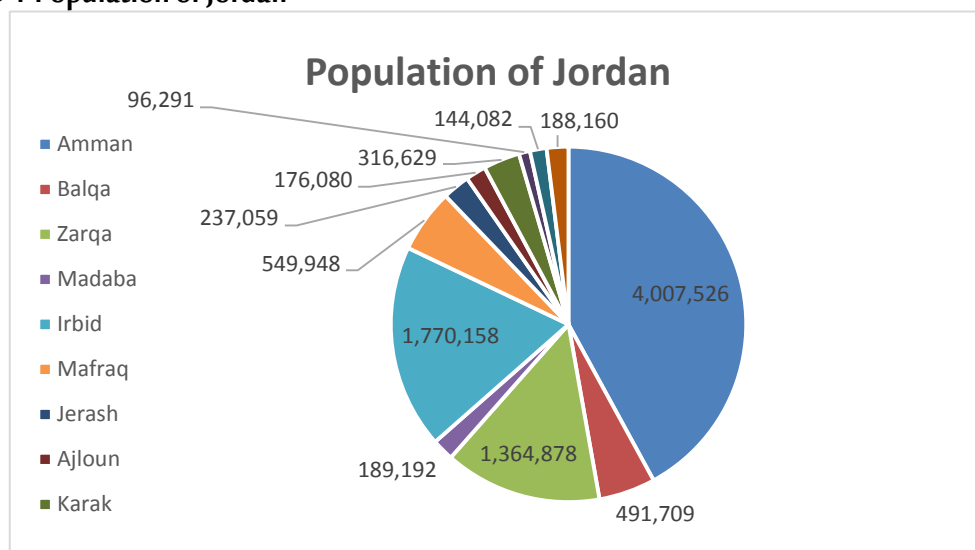
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year; figure below summarizes the population distribution among the kingdom governorates.

Figure 1 Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafrqa at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table1 : Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

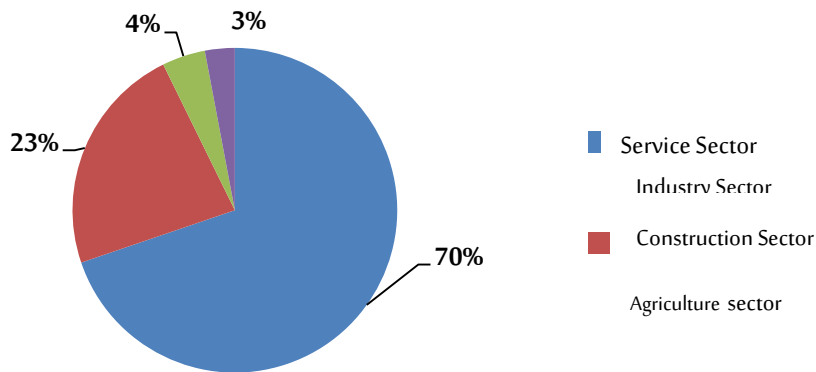
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure2 Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce. Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market Analysis

Project Description

The project is based on the idea of establishing a center for the collection and marketing of sheep milk in Karak particularly and can expand the activity to the neighboring governorates, which have large quantities of sheep milk.

Where the livestock breeders suffer in spring time, especially from their inability of marketing and distributing sheep milk, which leads to low self-sufficiency of milk. The problem of marketing and distributing milk in time, especially in the milking season, is one of the main problems faced by sheep and goat breeders in the Remote and desert areas. Due to their distance from consumers, this often leads to milk damage or to the decline of milk output or the exploitation of traders and contractors who work in the field of cheese industry the breeders in the desert areas and the purchase milk at unfair prices. Therefore, the establishment of a project for the collection, marketing and distribution of sheep and goat milk in remote areas will solve the problems experienced by livestock breeders in those areas. Such projects are projects that can easily control the size of the project, since most machinery and plant equipment can be available locally at the required shape and size. Therefore, in order to create the greatest number of jobs, the size of these factories is estimated to be between 5-6 tons of milk per day.

Project Objectives

The Jordanian market needs large amounts of this material, as the average value of imports of fresh and dried milk (HS 402) is about 80 million dinars for the years (2011-2015) of about 26 thousand tons.

1. Achieving a profitable return to the owner of the project,
2. Collection of milk from small livestock breeders
3. Marketing milk to dairy factories.
4. Creating job opportunities in the less developed regions,
5. Improve the economic and social conditions of sheep and goats breeders by purchasing their products.

The Proposed Product

Karak is famous for the industry of Jamid, which is often produced from sheep's milk. Breeders or small producers convert fresh milk to dairy products of al-Jamid and ghee in anticipation of milk damage. The project is based on the idea of establishing a specialized center in the collection of fresh milk and performing primary processes and keeping in refrigerated tanks and marketing to the dairy companies to produce dairy products from sheep's milk such as ewe's milk, ewe's yogurt, ewe's cheese and jamid. The

project will collect and cool the milk of ewes and goats by means of a refrigerated car specially prepared for this purpose from the sheep farms and then marketing and delivering them to the dairy factories in order to reduce the burden on the sheep breeders and increase their income.

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Market Size Analysis

The statistics of the department of statistics indicate that the quantities of imported liquid and dried milk amounted to 33 thousand tons have been imported from many countries; the most important are European Union countries and Saudi Arabia. Table 2 shows the value and quantity of imports of milk during the period (2011-2015), where the average quantity of imports of milk reached (25.9) thousand tons with an estimated value of (78 million JD). Table 2 shows that there is a deficit in the trade balance for the year 2014 amounted to about (71.2) million JD. Table (3) shows the local production of sheep milk, cow milk and milk consumption during the period (2011-2017). Where the percentage of self-sufficiency of milk about 41% in 2017, where the numbers and data of the Department of Statistics indicate that the production of sheep's milk in 2015 amounted to about (100) thousand tons.

Table 2: Value and quantity of imports, exports, re-exports and trade deficit of milk (HS 402)

Year	The value of Imports (000 JD)	Imports Quantity (Tons)	The Value of Local Exports (000 JD)	Local Export Quantity (Tons)	Re-exports Value (000 JD)	Re-exports Quantity (Tons)	Exports Total Value (000 JD)	Trade Balance Deficit
2011	62,851	19,871	4,619	861	4,411	2,205	9,030	-53,821
2012	77,536	25,727	78	17	7,077	2,947	7,154	-70,381
2013	69,193	22,428	67	86	6,573	2,527	6,640	-62,554
2014	93,788	28,379	0	0	5,923	2,141	5,923	-87,865
2015	88,939	33,244	0	0	7,362	3,826	7,362	-81,577
Avg	78,461	25,930	953	193	6,269	2,729	7,222	-71,240

Source: Department of Statistics (2017) Foreign Trade Reports

Local demand of fresh milk and dairy products was estimated based on the 2013 Household Expenditure Study, which estimated the average per capita consumption of various dairy products at 77 kg per year and fresh milk at 3.26 kg per year. By looking at the population census for 2015 and assuming population growth rate of 2.2% per year for the coming period, as indicated by the Department of Statistics, a population projection was carried out during the period (2017-2026) and then multiplied the expected population by the average per capita consumption of fresh milk and dairy products (Assuming constant consumption), the future demand of milk was estimated during the period (2017-2026) as shown in Table 3. The volume of the demand for the years 2017-2026 is expected to be as shown in Table 4, where the demand of fresh milk is expected to increase from 32,000 tonnes in 2017 to 39,000 tons in 2026.

Table 3 : The development of local production and consumption of milk during the period (2017-2026) Tons

The Size of the Jordanian Milk Market	2011	2012	2013	2014	2015	2016	2017*
Imports of fresh and dried milk (tonnes) (HS 402)	19,871	25,727	22,428	28,379	33,244	34,242	35,269
Exports (Tons) (HS 402)	861	17	86	0	0	0	0
Re-exported (Tons)	2,205	2,947	2,527	2,141	3,826	3,941	4,059
Local production of sheep milk (tonnes)	68,367	69,744	70,369	100,701	98,781	101,744	104,797
Local production of cow's milk (tonnes)	300,833	278,756	236,731	209,699	208,119	214,363	220,793
Local production of milk (2)	369,200	348,500	307,100	310,400	306,900	316,107	325,590
Consumption of milk (tons)	538,321	571,730	624,616	677,732	735,852	757,927	780,665
Import of powder milk (tons)	169,121	223,230	317,516	367,332	428,952	441,820	455,075

Source: Department of Statistics (2017), (2) Central Bank (2017)

Table 4 : Estimation of expected demand for fresh milk and powder milk during the period (2017-2026) Tons

Demand	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026
Expected population (1,000 people)	9,984	10,204	10,428	10,658	10,892	11,132	11,377	11,627	11,883	12,144
Annual per capita consumption (kg / person) powder milk	76.98	76.98	76.98	76.98	76.98	76.98	76.98	76.98	76.98	76.98
Annual per capita consumption (kg / person) Fresh milk	3.26	3.26	3.26	3.26	3.26	3.26	3.26	3.26	3.26	3.26
Local consumption of powder milk (ton) (1)	768,585	785,494	802,775	820,436	838,486	856,933	875,785	895,052	914,743	934,868
Domestic consumption of fresh milk (tonnes)	32,549	33,265	33,996	34,744	35,509	36,290	37,088	37,904	38,738	39,590

Competitors Analysis

The most important problem faced by livestock breeders in far-reaching areas is the speed damage of fresh milk and the lack of refrigerated transportation means to transport milk to the areas of consumption, as the produced quantity by the breeder does not justify the additional cost of providing refrigerated storage capacity and refrigerated transport vehicles. Therefore, breeders adopt the conversion of fresh milk to other products that can be stored for longer period, such as cheese, Labaneh, Jamid, butter and ghee in a rare productive conditions, and the cattle breeders in Karak governorate rely on the production of boiled cheese and Jamid. As their cattles spread far away from the areas of fresh milk, Labneh and yogurt consumption in and around the capital Amman. These products are sold at relatively high prices compared to imported products due to the higher quality of the local products and consumer preference for the local products rather than the imported one.

Fresh milk collectors are spread among breeders and buy fresh milk from them at low contractual prices to ensure a high profit margin for the supply of fresh milk either to their workshops or to be supplied to local cheese producers and Jamid. Ewes produce 60 kg of milk per ewe for about 120 days in milking season, which is half a kilogram of milk daily. Assuming that the herd consists of 400 heads of sheep and 70 of the herd are producing milk, (120-150) kg of milk per day on two morning and evening periods, and so on. It is worth to mention that milk of ewes is available throughout the year at varying rates, reaching peak in the spring season due to the health care and good nutrition of the herd as well as good management, Which leads to three births of asheep every two years which means three seasons of milking / two years, as is the case for goats, which have a period of milking, extends to 150 days, with a higher percentage of births. So large quantities of fresh milk are produced and breeders find some difficulty in marketing, especially in the spring periods, forcing some of them to leave the babies breastfeed ed from their mothers because there are no centers for the collection or marketing of milk in the desert or rural areas or areas that take care of sheep. Cooperative societies concerned with raising sheep can adopt such projects.

The competition for the project is limited to those working in the field of cheese processing in remote areas or the collection and marketing of milk to the milk factories, who are traders and contractors practicing this profession. Based on available information, no milk collection center is available in the proposed project area in Al-Karak governorate. The main competitors are producers of locall Jamid in Karak governorate; the Shafa al-Khair Association, the Karak Livestock Breeding Society, the Adar Factory, Mansheyet Abu Hamour Society, and the Zohum Jamid and Dairy Products Factory.

Therefore, it is not expected that the project may face difficulties in achieving the expected sales, considering that sheep and goats' milk are locally desirable food commodities for their nutritional importance. Jordan still needs large quantities of milk, either for direct consumption in the form of fresh milk or processed in the form of other derivatives. It also imports fresh milk and long-term milk from Saudi Arabia and neighboring countries to cover part of the local market need.

Prices Analysis and Pricing Policy

The fresh milk from sheep are sold directly to some dairy factories in the nearby areas or are sold to milk collectors who, in turn, produce it into different types of dairy products and sell them in urban areas for 1 JD perone kg in times of drought (dry ewes before and during pregnancy) and (0.6) dinars during the spring season, the traders of milk produce many dairy products from it machine to sell the product to the consumer in the form of dairy and fresh milk at a rating between (1.75-1.80 JD) for one Kg of sheep's milk in the form of pasteurized milk or fresh daires. There is always a complaint from milk producers concerning the low profit margin compared to the prices paid by the consumer for retailers and dairy plants.

The project aims to collect from breeders milk of high quality, free of impurities and antibiotics and in accordance with the Jordanian standards milk from breeders, approved to be free of hair and dust ; whereas the milk of the original ewes and goats milk should be without mixing with cows milk to carry out the collection, storage and cooled transport, it's suggested that the purchase price is rewarding for the breeders to ensure Long-term contracting and ensuring continuity of supply at (0.75) JD per kg throughout the season regardless of quantity or condition of pasture and seasonal and weather conditions. It should be cooled and transfered to dairy factories at a contractual price of (1.00) JD per kg.

The milk of sheep is characterized by high fat and good taste, but there are no regular large quantities of sheep's milk to be supplied to milk factories on a daily basis. As for cow's milk, many dairy factories have been shown through civil visits or telephone calls to their desire to increase the manufacturing capacity of their factories of sheep's milk, providing them with the required quantities on a daily basis, and to set a fixed price throughout the year to be agreed each year.

Market Share

The project's market share is about 750 tons per year of fresh milk determined by the assumption that the center will collect 5 tons of fresh milk daily from Karak governorate along the milking season which is estimated by 150 days and may extend to longer periods during the year. Or that the supplied quantities will be decreased gradually with the entry of the season of mating and pregnancy for sheep, which starts usually from June to the beginning of December and starts with it the birth season. One month is left for breastfeeding and after that the obtained amount of fresh milk will be half of the normal quantity. After two months, the weaning phase begins, then full quantities of produced milk can be obtained from ewes. The estimation of quantities of the production of sheep milk in Al-Karak governorate was left to ascertain the availability of raw materials to be supplied to the plant, Where the data of the Directorate of Animal Production in the Ministry of Agriculture indicates that the number of sheep in Al-Karak governorate and its districts amounted to 607 thousand heads at the end of 2015, constituting about 14.3% of the total number of sheep in the Kingdom, producing 15,742 tons of sheep milk (sheep and goats) annually. Thus, the required quantity needed of sheep milk for the project is (750 tons) that does not constitute more than 5% of the production capacity of Al-Karak. Therefore, there is no fear of the lack of the required quantities of milk for the project from Al-Karak governorate.

Table 5 : Estimation of Al-Karak Sheep Milk in Tons

District	Number of sheep (Head)	Number of Goats (Head)	Total Number (Head)	Sheep Milk (Ton)	Goat Milk (Ton)	Total Milk Quantity (Ton)
Al-Qasabah	107,028	34,253	141,281	2,246	1,388	3,660
Al-Aghwar Al-Janubiah	24,783	52,261	77,045	520	2,118	1,996
Al-Mazar Al-Janoubi	117,226	33,847	151,073	2,460	1,372	3,914
Al-Quttranieh	106,079	15,828	121,907	2,226	642	3,158
Al-Qaser	42,400	19,286	61,687	890	782	1,598
Faqou'	31,020	12,093	43,114	651	490	1,117
A'ie	6,081	5,443	11,524	128	221	299
Governorate Total	434,618	173,013	607,631	9,122	7,012	15,742
Kingdom Total	3,173,658	1,067,557	4,241,215	66,609	43,268	109,877

It is possible to estimate the project's market share based on the expected production capacity of the project, which amounts to 4.8% of the production of Al-Karak Governorate or about 0.68% of the local production.

Projected Revenues Analysis

Sheep and goat milk can be produced throughout the year in the case of the availability of sheep milk, and since there is a period of surplus and periods where there is a shortage of sheep milk, the center can produce other dairy products such as yogurt, soft Labaneh and white cheese so that The center won't stop production. But for the purposes of the study, the focus is on the collection of sheep milk from different districts of Al-Karak and marketing the milk to dairy factories in other governorates, especially Amman and Zarqa, where most factories are located. Table (6) shows the projected sales of the project. For the purposes of this study, it was considered that the average milk selling price (1.00) JD/kg and the price of purchasing from breeders (0.75) JD/Kg throughout the year and it is expected to increase by 2% per year.

Table 6 : The Projected Revenues (2018-2027)

Project Revenues	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total Sales (Ton)	375	750	765	780	796	812	828	845	862	879
The weighted price of the commodity (JD / ton)	1,000	1,020	1,040	1,061	1,082	1,104	1,126	1,149	1,172	1,195
Total sales (thousand dinars)	375.0	765.0	795.9	828.1	861.5	896.3	932.5	970.2	1,009	1,050.2

4. Techniacal Analysis

Any collected quantity from remote and desert areas and supplied to dairy factories will lead to a reduction in imports of milk and dairy products. Sheep breeders in remote areas do not fully exploit the quantities of milk produced, except for the manufacture of low amounts of cheese, butter and ghee, mostly for household consumption. The rest of the milk is left to feed the young sheep for a long time throughout the milking season. The milk collection center can start with the capacity of about 5 tons of milk per day. Note that some traders and contractors practice the profession of milk collection in primitive ways. Noting that the amount of milk supplied to dairy factories (Al-Ra'i, Al-Sanabel, Hammouda, Masoud, Biladi, Jordanian Danish, Shami, Al-Dyiar, Al-Mouruj and Al-Badyiah), based on the estimation of the owner of the Al-Ra'i dairy factory are about 25 tons perday, and most of them are collected from Mafraq area. Dairy plants are able to a 100 tons per day of sheep's milk.

Al Awassi sheep milk will be collected from remote areas and cooled until marketed to dairy factories in the same day under the following conditions:

- The milk shall be received or transferred by the competent employee within two hours of the milking time by a car prepared for this purpose, which should be refrigerated and the receipt of milk should be done twice daily (6 am and 5 pm). The center shall determine specific times for the receipt of milk from the breeders who wish to supply the milk themselves within the agreed terms, and shall not receive any quantity from them after the specified time.
- Milk amounts must be free from any antibiotics or acidity.
- The milk collection center should set dates for the receipt of milk from livestock breeders

The collection of milk in remote areas is difficult because of the movement of herds of sheep from one location to another, the seasonal production of sheep and the small quantities of milk produced for each breeder. However, the availability of the road network at proposed sites such as Al-Karak, water and electricity networks has brought many sheep breeders into one area, making it easier to collect milk.

The milk collection center consists of refrigerated transport vehicles, a unit for receiving milk, weighting and cleaning, and then stored in refrigerated tanks. The production units can be described as follows:

- Department of milk receipt, weighing and testing , in addition to cleaning unit section includes a balance, milk refineries, pumps and heat exchanger
- Treatment and cooling section

- Department of power equipment (steam, electricity, air conditioning and refrigeration) as well as cleaning units.
- The maximum estimated capacity of the plant was at 5 tons per day; 750 tons per year, with one shift daily and 300 working days per year. It has been assumed that the project will be operated by utilizing 50% of its estimated capacity in the first year and that the quantities of production will increase to the full capacity in the second year.

Raw Materials Requirements

It is necessary to purchase 750,375 kg of fresh milk as a small percentage of milk will be lost and damaged at a purchase price of JD 0.75 per kilogram. The price of sheep's milk starts at JD1 at the beginning of the season (January) and reduced to (0.6 JD) at the peak of the milking season (March, April and May). The average price of fresh milk from the breeders was assumed by (0.75 JD), where the center will take charge of the transportation and collection process. The main raw materials, as well as sheep's milk, are the disinfectants and chemicals needed for the laboratory and the workers' supplies of uniform and toiletries. Table 7 shows the cost of raw materials, fresh milk and the annual cost of marketing 750 tons of milk.

Table 7 : Prices, Quantities and Cost of the Annual Raw Material

Raw Material	Price (JD/Kg)	Quantity	Value
Fresh Sheep Milk	0.75	750,375	562,781
Chemicals and Disinfectants	1.00	1,200	1,200
Total			563,981

The Required Human Resources

This project needs an agricultural engineer specialized in food processing in addition to a delegate who handles the supply of milk to the factory, marketing the product to the target markets, drivers to bring milk and guards for the center. Table (8) shows employment needs and their approximate wages:

Table 8 : Required Table

Job Title	No.
Indirect Employees	
Agricultural Engineer/Food Processing	1
Supply and Marketing Employee	1
Guard	1

Job Title	No.
Total Indirect Employees	3
Laboratory Employee	1
Drivers	4
Cleaner and reporter	1
Total Direct Employees	6
Total Employees	9

Job Description

Agricultural Engineer / Food Processing: Preparation of the production plan for the receipt and marketing of fresh milk, identification and schedule of supply and sale, supervision of the process of processing the initial milk and cooling and storing in refrigerated tanks. Follow up the process of receiving milk, weight, and check its validity, and matching specifications. Supervising the status of milk in the center's equipment. Evaluating the performance of technical staff, identifying their training needs, training them and developing their skills. Mobilization of business models. Receiving, preparing and matching the specifications of the pasteurized milk; Preparing, operating and monitoring the dairy products and machines; supervising cleaning of the machines and equipment; Filling work forms; and Following the procedures and regulations of occupational safety and health.

Marketing and Supply representative: Development of plans for the production, sale and marketing of products, supervision the procurement of milk quantities to made the product, selection of good herds of small ruminant breeders, control of records of production and sales, revenues and expenses. Monitoring and developing quality systems and standards. Ensure sufficient supply of the raw material requirements, production, and sale. Development of regulatory procedures and instructions to monitor the receipt of fresh milk and the inspection and matching quantities. Contact dairy factories and dairy industries to coordinate the requests for sale, and fill forms for the process of receipt and delivery, and follow the process of loading and unloading and receipt, bargaining on the prices of sales and purchase of raw milk, Respond to customer inquiries and secure their requests. Invoice editing, inventories of warehouse contents, organization of sales records and financial records, filling of business models and application of occupational and public safety and health procedures and instructions.

Production technician and laboratory supervisor: Supervise the receipt of fresh milk containing high-fat, and take a sample to examine the proportion of fat. Control of chemical and biological contamination, sampling of each supplier, and testing to ensure that the product conforms to the required specifications such as pH and fat content, free of impurities and antibiotics and follow-up supervision of cleaning the center's equipment, tools and temperature. Laboratory of samples of quality specifications, filling of working forms. Application procedures and instructions of the occupational safety and health.

The driver of milk production center: works under the supervision of the agricultural engineer. Receives the milk from herders, transport it by refrigerated cars and bears to receive and weigh milk from the places of the presence of sheep breeders, weight of milk, and check the validity, and conformity of specifications, apply the procedures and instructions for safety and health professional and public

Site Analysis

Based on all these data, it is proposed that a milk collection center should be established in Al-Karak governorate so that the appropriate location for production is selected based on the availability of infrastructure services such as water and electricity, availability of raw materials and the prevailing production system. For example, As a result of the presence of large numbers of sheep (Table 9) and the availability of raw material, but the critical point in these projects is the provision of refrigerated transportation to transport milk from the breeders to the center, and can be done using small mobile refrigerators to transport milk to the project site. During spring season from January to April and July.

Table9 : Percentage and Number of Sheep in Karak Governorate by Administrative Divisions

Districts	Number of Livestock			Percentage		
	Sheep	Goats	Total Livestock	Sheep	Goats	Livestock
Al-Qasabah	107,028	34,253	141,281	3.4%	3.2%	3.3%
Al-Aghwar Al-Janubiyah	24,783	52,261	77,045	0.8%	4.9%	1.8%
Al-Mazar Al-Janoubi	117,226	33,847	151,073	3.7%	3.2%	3.6%
Al-Quttranieh	106,079	15,828	121,907	3.3%	1.5%	2.9%
Al-Qaser	42,400	19,286	61,687	1.3%	1.8%	1.5%
Faqou'	31,020	12,093	43,114	1.0%	1.1%	1.0%
A'ie	6,081	5,443	11,524	0.2%	0.5%	0.3%

Governorate Total	434,618	173,013	607,631	13.7%	16.2%	14.3%
Kingdom Total	3,173,658	1,067,557	4,241,215	100.00%	100.00%	100.00%

Description of the Production Process and Manufacturing Steps

The center consists of several sections, the most important are:

1. **Department of receiving and storage of raw milk refrigerated:** This section examines the acidity of the milk, which determines the acceptance or rejection of milk, and then the weight of milk and refrigerated and stored in large tanks until the time of manufacturing process.
2. **The storage department:** consists of several large rooms to store refrigerated milk in stainless steel tanks, refrigerated until the milk is to be transformed, to dairy factories by special cars and tanks of transportation.
3. **Laboratory Department:** equipped with all the devices and tools and materials necessary for the examination of raw milk and evaluation, and determine the quality of fat and the extent of conformity to standard specifications.
4. **Management, marketing and various other services Department:** It includes management offices and public facilities of restaurants, bathrooms and a hall for sale.

Technical Requirements Analysis

The Required Land

The project needs a total area of (1000) square meters and 450 square meters allocated for building, reception and storage, and the remaining space is allocated for services, facilities and aisles.

Construction Works

The project needs the following buildings and facilities: management office, milk collection and storage hall, refrigerated warehouses. The cost of the construction works required for the project includes the design of the building, the necessary schemes of the project, Paving and excavation works garages.

Shown in (Table 11) as follows:

Table 10 : Required Construction Work

Construction Works	Area (m ²)
Administrative Offices	100
Laboratory	50
Collection, Receiving and Storage Hall	100
Refrigerated Warehouses	100

Paving and Site Preparations 1,000

Machinery and Equipment

Table (11) shows the needs of the project of machinery and equipment for the production of Jamid, butter and ghee

Table 11 : Required Machinery and Equipment

Machinery and Equipment	Quantity
Receiving Tank and Balance	1
Scissor Cooler	1
Stainless Steel Cooling Containers (1 Ton)	2
Stainless Steel Cooling Containers (2 Ton)	2
Milk Containers	250
Pipes	1
Electricity Generator	1
Special Equipment for Cooling Room	1
Others	1
Laboratory Tools	1
Other Tools	-

Laboratory tools include sensitive balances, water bath, centrifuge, lactometer, PH meter, glassware, chemicals and other necessary tools to check for bacterial contamination and quality of production.

Furniture

The project needs simple furniture for management and permanent employment estimated at 3000 dinars, which are simple tables and offices, cabinets, laboratory furniture and workers' rest.

Transportation and Vehicles

The project needs refrigerated transport vehicles to transport milk from the area of the breeders to the plant as well as to a medium transport vehicle to market the production to dairy factories as shown in (Table 12)

Table 12 : Required Transportation and vehicles

Transportation and Vehicles	Quantity
Refrigerated Milk transfer Truck	4
Refrigerated Milk Tank	1

Essential Services

Electricity, water and transportation services are available in most parts of the Kingdom and are assumed to be available on the project site. The essential needs of the project (Utilities, Electricity, Water, Diesel, Fuel and Steam) are estimated to be about 2% of the total annual revenues as shown in Table 13

Table13 : Required Essential Services

Expenses Assumptions	
Service Utilities Expenses from the Revenues	0.50%
Maintenance Expenses from the Revenues	0.50%
Consumers Expenses from the Revenues	0.50%
Insurance Expenses from the Assets Value	0.15%
Marketing Expenses from the Revenues	1.00%

Legal Requirements and Procedures

The dairy industry needs licenses and constant and strict control, but the fact that the project does not seek to produce food for consumers, but the commodity to be traded is an inputs for other industries which is dairy industry, so before the start of the investment. Many legal and administrative conditions should be achieved. These include the registration of a legal establishment, the obtaining of necessary licenses and approvals, etc. Delivery of required documents and payment of the initial administrative costs associated with the investment.

- 1- Legal registration whether it is an individual institution or a company (solidarity, simple recommendation, private contribution, Limited Liability Company, etc.)
- 2- Registration of the trade name of the company requesting the registration of the trade name.
- 3- Registration at the Chamber of Industry / Commerce. Every establishment engaged in industrial or commercial activity must be registered in the Chamber of Industry / Commerce.
- 4- Professional licensing - a prerequisite for legal entities to be able to carry out business activities.
- 5- Registration at the General Organization for Social Security - Every legal establishment is bound by the provisions of Jordanian law to register for social security.
- 6- FDA approval is not required because the product is not intended for consumers.

5. Financial Analysis

Assumptions

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Calculation Assumptions:

- The weighted average cost of capital used in the study is 11.78% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Income tax on the net project income is 5 % during the life time of the project.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Table 14: Annual Growth Rates Assumptions

	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	2.00%
Annual Expenses Increase Rate	2.00%
Annual Increase Rate in Employees Salaries	3.0%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Capital Expenditures

The estimated capital costs of the project are 386,724 JD and cover the cost of land, construction, machinery and equipment, furniture and furnishings, vehicles and transportation, pre-operating expenses and initial working capital. These costs are shown in the following tables:

Table 15: Summary of expenses

Summary of capital expenditures	Cost (JD)	Contingency	Cost (JD)	Percentage
Land	25,000	10%	27,500	%7.1
Construction Works	57,250	10%	62,975	%16.3
Machinery and Equipment	53,350	10%	58,685	%15.2
Furniture	3,000	10%	3,300	%0.9
Vehicles and Transportation	110,000	10%	121,000	%31.3
Pre-operating expenses	4,500	10%	4,950	%1.3
Initial working capital	98,467	10%	108,314	%28.0
Total (JD)	351,567		386,724	100%

Table 16: Construction Work Capital Expenditure

Construction Works	Cost (JOD per Squared Meters)	Area (Squared Meters)	Gross Cost / Jordan Dinar
Offices	150	100	15,000
Laboratory	150	50	7,500
Warehouse	150	100	15,000
Cooled stores	100	100	10,000
Site processing	5	1,000	5,000
Others	-	-	4,750
Gross Total			57,250

Table 17: Machinery and equipment Capital Expenditure

Machinery and equipment	Quantity	Unit cost	Gross Cost/ Jordan Dinar
Receiving Tank and Balance	1	10,000	10,000
Scissor Cooler	1	3,000	3,000
Stainless Steel Cooling Containers (1 Ton)	2	3,000	6,000
Stainless Steel Cooling Containers (2 Ton)	2	4,000	8,000
Milk Containers	250	50	12,500
Pipes	1	500	500
Electricity Generator	1	4,000	4,000
Special Equipment for Cooling Room	1	2,500	2,500
Others	1	1,000	1,000
Laboratory Tools	1	1,000	1,000
Other Tools	1	-	4,850
Gross Total			53,350

Table 18: Furnitutre Capital Expenditure

Furnitutre	Quantity	Unit cost	Gross Cost /Jordan Dinar
Offices furniture	1	2,000	2,000
Laboratory equipment	1	1,000	1,000
Gross Total			3,000

Table 19: Transporation and Vehicles Capital Expenditure

Transporation and Vehicles	Quantity	Unit Cost/JOD	Total Cost/JOD
Refrigerated Milk transfer Truck	4	15,000	60,000
Refrigerated Milk Tank	1	50,000	50,000
Gross Total	5		110,000

Table 20: Pre-operating expenses

Pre-operating expenses	Gross Cost
Governmental fees	500
Transportation	1000
Marketing	1000
Training	2000
Total	4,500

Table 21: Initial working capital

Initial working capital	Number of months	Percentage	Cost (JD)
Operating expenses	3	%25	91,218
Administrative and general expenses	3	%25	512
Marketing expenses	3	%25	938
Indirect salaries	3	%25	5,800
Gross Total			98,467

Depreciations

:The following tables show all equipment costs, expenses and depreciation rates

Table 22: Capex and Depreciation Expenses

Costs of equipment, capital expenditures and annual depreciation rates	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	27,500	%0.0	%0.0
Construction Works	62,975	%5.0	%0.0
Machinaries	58,685	%10.0	%2.0
Furniture	3,300	%10.0	%5.0
Vehicles	121,000	%10.0	%0.0
	273,460		

Sources of funding

The financing structure required for the project consists of loans and equity. The investment cost of this project was estimated at JD 445,710 JD. This amount is determined through fixed assets, which include land, construction, machinery and equipment, furniture and furnishings, vehicles and transport vehicles, pre-operating expenses and initial working capital. (50%) of the project will be financed through loans (222,855 JD) and the rest (50%) through self financing (equity). The following table shows the general structure of the required financing:

Table 23: Depreciation and annual increases on construction work

Summary of capital costs, additions and annual depreciation	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total fixed assets	273,460	274,799	276,169	277,572	279,009	280,480	281,986	283,529	285,109	286,728
Total consumption expenses	21,447	21,581	21,718	21,858	22,002	22,149	22,300	22,454	22,612	22,774
Total accumulated depreciation	21,447	43,028	64,747	86,605	108,607	130,756	153,056	175,510	198,123	220,897
Total additions	0	1,339	1,370	1,403	1,437	1,471	1,506	1,543	1,580	1,619
Total net book value	252,013	231,770	211,423	190,967	170,402	149,723	128,930	108,019	86,987	65,832

Table 24: Sources and use of finance

Sources of funding	amount	Percentage
Equity (self-financing)	193,362	%50
Loans	193,362	%50
Gross Total	386,724	%100
Use of funding	amount	Percentage
Capital expenditure	273,460	%70.71
Pre-operating expenses	4,950	%1.28
Working capital	108,314	%28.01
Gross Total	386,724	%100.00

Table 25: Loan details

	Unit	Value
Loans	Percentage	50%
Equity	Percentage	50%
Annual Interest Rate	Percentage	9.50%
Loans Period	Year	6
Loan Starts at year:	Year	2018
Grace Period	Year	1

Annual Revenues

The price of fresh milk was estimated at 1000 dinars per ton, or 1 dinar per kilogram, and is increased by 2 percent annually, as indicated in the market study, based on the prevailing milk prices imported to the dairy factories, the price should be very competitive as the product will compete with current producers in the local market. As shown in (Table 26)

Operating expenses

Table (27) shows the cost details of the cost of milk purchased, chemicals and materials.

Administrative and general expenses

Table (28) shows the expected administrative and general expenses for the proposed project (2018-2027). Table (29) shows the expected marketing expenses of the proposed project. Marketing expenses account for 1% of revenues.

Human Resources

The salaries and monthly wages of each employee were estimated to be multiplied by 16 months of the year for the purpose of calculating the increase in salaries and annual wages due to social security, health insurance, etc. to the employees of the administration. The following table shows the expected annual salaries and wages. The annual rate of increase in wages and salaries is taken into account:

Table 26: Revenues of the project during the years 2018-2027

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total sales (tons)	375	750	765	780	796	812	828	845	862	879
Item Price (JD / ton)	1,000	1,020	1,040	1,061	1,082	1,104	1,126	1,149	1,172	1,195
Total sales (JD)	375,000	765,000	795,906	828,061	861,514	896,319	932,531	970,205	1,009,401	1,050,181
Total Project Income (JD)	375,000	765,000	795,906	828,061	861,514	896,319	932,531	970,205	1,009,401	1,050,181

Table 27: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw material expenses	281,991	563,981	575,261	586,766	598,501	610,471	622,681	635,134	647,837	660,794
Service expenses	1,875	3,825	3,980	4,140	4,308	4,482	4,663	4,851	5,047	5,251
Maintenance expenses	0	3,825	3,980	4,140	4,308	4,482	4,663	4,851	5,047	5,251
Consumables	1,875	3,825	3,980	4,140	4,308	4,482	4,663	4,851	5,047	5,251
Direct salaries	563	1,148	1,194	1,242	1,292	1,344	1,399	1,455	1,514	1,575
Other expenses	45,400	30,488	31,403	32,345	33,315	34,315	35,344	36,404	37,496	38,621
Gross Total	33,170	60,709	61,980	63,277	64,603	65,958	67,341	68,755	70,199	71,674
Raw material expenses	364,873	667,801	681,776	696,051	710,635	725,533	740,753	756,302	772,188	788,418

Table 28: administrative and general expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Communications expenses	500	510	520	531	541	552	563	574	586	598
Hospitality expenses	250	255	260	265	271	276	282	287	293	299
Stationery and prints	200	204	208	212	216	221	225	230	234	239
Insurance expenses	410	418	427	435	444	453	462	471	481	490
Other expenses	186	139	142	144	147	150	153	156	159	163
Gross Total	2,046	1,526	1,557	1,588	1,620	1,652	1,685	1,719	1,753	1,788

Table 29: Marketing Expenses

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	3,750	7,650	7,803	7,959	8,118	8,281	8,446	8,615	8,787	8,963

Table 30: Expected yearly salaries for the years (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
Agricultural Engineer	12,800	13,184	13,580	13,987	14,407	14,839	15,284	15,742	16,215	16,701
Marketing and supply officer	6,400	6,592	6,790	6,993	7,203	7,419	7,642	7,871	8,107	8,351
Guard	4,000	4,120	4,244	4,371	4,502	4,637	4,776	4,919	5,067	5,219
Total indirect salaries	23,200	23,896	24,613	25,351	26,112	26,895	27,702	28,533	29,389	30,271
Direct Employees										
Laboratory Worker	6,400	6,592	6,790	6,993	7,203	7,419	7,642	7,871	8,107	8,351
Drivers	19,200	19,776	20,369	20,980	21,610	22,258	22,926	23,614	24,322	25,052
Cleaners	4,000	4,120	4,244	4,371	4,502	4,637	4,776	4,919	5,067	5,219
Total direct salaries	29,600	30,488	31,403	32,345	33,315	34,315	35,344	36,404	37,496	38,621
Gross Total	52,800	54,384	56,016	57,696	59,427	61,210	63,046	64,937	66,885	68,892

Income Statement

Table 31: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	375,000	765,000	795,906	828,061	861,514	896,319	932,531	970,205	1,009,401	1,050,181
Gross Operating Revenues	375,000	765,000	795,906	828,061	861,514	896,319	932,531	970,205	1,009,401	1,050,181
Operating Expenses	(364,873)	(667,801)	(681,776)	(696,051)	(710,635)	(725,533)	(740,753)	(756,302)	(772,188)	(788,418)
Gross Operating Profit	10,127	97,199	114,130	132,009	150,880	170,787	191,778	213,903	237,214	261,764
Gross Profit Percentage	%3	%13	%14	%16	%18	%19	%21	%22	%24	%25
Salaries and Benefits (Indirect Staff)	(23,200)	(23,896)	(24,613)	(25,351)	(26,112)	(26,895)	(27,702)	(28,533)	(29,389)	(30,271)
General and Administrative Expenses	(2,046)	(1,526)	(1,557)	(1,588)	(1,620)	(1,652)	(1,685)	(1,719)	(1,753)	(1,788)
Marketing Expenses	(3,750)	(7,650)	(7,803)	(7,959)	(8,118)	(8,281)	(8,446)	(8,615)	(8,787)	(8,963)
Pre Operating Expenses	(4,950)									
Gross Indirect Expenses	(33,946)	(33,072)	(33,973)	(34,898)	(35,850)	(36,828)	(37,833)	(38,867)	(39,930)	(41,022)
Income before Interest, Depreciation and Tax	(23,820)	64,127	80,158	97,111	115,030	133,959	153,945	175,036	197,284	220,742
Fixed Assets Depreciations	(21,447)	(21,581)	(21,718)	(21,858)	(22,002)	(22,149)	(22,300)	(22,454)	(22,612)	(22,774)
Income before Tax and Interests	(45,267)	42,546	58,440	75,253	93,028	111,810	131,645	152,582	174,672	197,967
Bank Interests	(18,369)	(15,330)	(12,003)	(8,359)	(4,369)	0	0	0	0	0
Income Before Tax	(63,636)	27,215	46,437	66,894	88,659	111,810	131,645	152,582	174,672	197,967
Income Tax	0	0	(1,267)	(3,345)	(4,433)	(5,590)	(6,582)	(7,629)	(8,734)	(9,898)

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Profit	(63,636)	27,215	45,170	63,549	84,226	106,219	125,063	144,953	165,938	188,069
Net Profit Percentage	%17-	%4	%6	%8	%10	%12	%13	%15	%16	%18
Compulsory Reserves	0	(2,722)	(4,517)	(6,355)	(8,423)	(10,622)	(12,506)	(14,495)	(16,594)	(18,807)
Retained Earnings	(63,636)	(39,142)	1,510	58,705	134,508	230,105	342,662	473,119	622,464	791,726
Expected Balance Sheet										

Table 32: Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	(23,438)	(76,031)	(54,022)	(17,377)	35,849	156,944	296,764	456,348	636,785	839,211
Inventory	23	47	48	49	50	51	52	53	54	55
accounts receivable	62,500	127,500	132,651	138,010	143,586	149,387	155,422	161,701	168,234	175,030
Total assets traded	39,086	51,516	78,677	120,682	179,484	306,382	452,238	618,102	805,072	1,014,297
Non-current assets										
Fixed assets (net)	252,013	231,770	211,423	190,967	170,402	149,723	128,930	108,019	86,987	65,832
Total non-current assets	252,013	231,770	211,423	190,967	170,402	149,723	128,930	108,019	86,987	65,832
Total assets	291,099	283,286	290,100	311,650	349,886	456,105	581,168	726,121	892,059	1,080,128
Liabilities										
Current liabilities										
Payables	0	0	0	0	0	0	0	0	0	0
Remaining amount of Loan	35,028	38,356	42,000	45,990	0	0	0	0	0	0

Total current liabilities	35,028	38,356	42,000	45,990	0	0	0	0	0	0
Non-current liabilities										
long term loans	126,345	87,989	45,990	0	0	0	0	0	0	0
Total long - term liabilities	126,345	87,989	45,990	0	0	0	0	0	0	0
Total liabilities	161,373	126,345	87,989	45,990	0	0	0	0	0	0
Property rights										
Members Contributions	193,362	193,362	193,362	193,362	193,362	193,362	193,362	193,362	193,362	193,362
Compulsory reserve	0	2,722	7,239	13,593	22,016	32,638	45,144	59,640	76,233	95,040
Profit (loss)	(63,636)	(39,142)	1,510	58,705	134,508	230,105	342,662	473,119	622,464	791,726
Total equity	129,726	156,941	202,111	265,660	349,886	456,105	581,168	726,121	892,059	1,080,128
Total liabilities and equity	291,099	283,286	290,100	311,650	349,886	456,105	581,168	726,121	892,059	1,080,128

Expected Cashflows Statement

Table 33: Expected Cashflows Statement

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	(63,636)	27,215	45,170	63,549	84,226	106,219	125,063	144,953	165,938	188,069
Non-monetary adjustments										
Bank Interests	18,369	15,330	12,003	8,359	4,369	0	0	0	0	0
Depreciation	21,447	21,581	21,718	21,858	22,002	22,149	22,300	22,454	22,612	22,774
Total Operating Cashflows before Additions to Working Capital	(23,820)	64,127	78,891	93,767	110,597	128,368	147,363	167,407	188,551	210,843
Inventory (Increase/Decrease)	(23)	(23)	(1)	(1)	(1)	(1)	(1)	(1)	(1)	(1)
Accounts Receivables (Increase/Decrease)	(62,500)	(65,000)	(5,151)	(5,359)	(5,576)	(5,801)	(6,035)	(6,279)	(6,533)	(6,797)
Accounts Payables (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Working Capital (Increase/Decrease)	(62,523)	(65,023)	(5,152)	(5,360)	(5,577)	(5,802)	(6,036)	(6,280)	(6,534)	(6,798)
Net Cashflows from Operating Activities	(86,343)	(897)	73,739	88,407	105,021	122,567	141,326	161,127	182,017	204,045
Cashflows from Investments Activities										
Fixed Assets (Procurement)	(273,460)	(1,339)	(1,370)	(1,403)	(1,437)	(1,471)	(1,506)	(1,543)	(1,580)	(1,619)

Net Cashflows from Investments	(273,460)	(1,339)	(1,370)	(1,403)	(1,437)	(1,471)	(1,506)	(1,543)	(1,580)	(1,619)
Activities										
Cashflows from Financing Activities										
Capital	193,362									
Loan Amortization	(31,989)	(35,028)	(38,356)	(42,000)	(45,990)	0	0	0	0	0
Bank Interest Rate	(18,369)	(15,330)	(12,003)	(8,359)	(4,369)	0	0	0	0	0
Loans	193,362	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing	336,366	(50,359)	(50,359)	(50,359)	(50,359)	0	0	0	0	0
Activities										
Net increase (decrease) in cash	(23,438)	(52,594)	22,010	36,645	53,225	121,096	139,820	159,584	180,436	202,426
Cashflows at the Beginning of Period	0	(23,438)	(76,031)	(54,022)	(17,377)	35,849	156,944	296,764	456,348	636,785
Cashflows at the end of Period	(23,438)	(76,031)	(54,022)	(17,377)	35,849	156,944	296,764	456,348	636,785	839,211

Feasibility Indicators

In order to access the discounted cash flow statement, the discount factor must be calculated where there is more than one method used to calculate it, but in the study period the weighted average cost of capital (WACC) is used as a discount factor to calculate the discounted cash flows and net present value of the investment. The financing structure of the project is based on equity and loans. Therefore, the discount rate should be adjusted according to the weighted average cost of capital (WACC). This is based on the assumptions shown in Table (34):

Table 34: Risk Premium

	Rate
(1) Risk Free Rate of Return	5.80%
(2) Return to Debt Maturity	7.50%
(3) Market Risk Premium	9.93%
Income Tax Rate	5.00%
(4) Beta	1
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 35: Weighted Average Cost of Capital

Weighted Average Cost of Capital	Indicator
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%7.50
Market Risk Premium	%9.93
Income Tax Rate	%5
Bets	1.00
Equity	193,362
Loans Value	193,362
Loans	
Cost of Borrowing before Tax	%7.50
Cost of Borrowing After Tax	%7.13
Loans Value	193,362

Loans Percentage	%50
Equity	
Cost of Equity	%16.4
Equity Value	193,362
Equity Percentage	%50
Gross	
Project Value	386,724
Weighted Average Cost of Capital	%100
WACC	%11.78

The following table illustrates the net expected free cash flow of the project:

Table 36: Free Net Cash flows Table

Net Cashflows	Pre- Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	(386,724)	(81,393)	(2,235)	72,368	87,004	103,584	121,096	139,820	159,584	180,436	202,426	2,706,828
Discount Factor	1.00	0.89	0.80	0.72	0.64	0.57	0.51	0.46	0.41	0.37	0.33	0.33
Net Present Value for Cash Flows	(386,724)	(72,817)	(1,789)	51,818	55,734	59,364	62,087	64,134	65,487	66,242	66,485	889,031

Table 37: Financial Analysis Results

Weighted Average Cost of Capital (WACC)	Rate	%11.78
Net Present Value for Cashflows	JD	919,052
Payback Period	Year	6
Internal Rate of Return	Rate	%25.64

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Table 38: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%25.64	6	%11.78
Revenues declined by 10%	%12.53	10	%11.78
Operating Expenses Increased by 10%	%15.78	10	%11.78

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 919,052 Jordan Dinars considering that the project provides 9 Job Opportunities for the governorate residents.